SVM - 1

Wednesday, 5 November 2025 7:50 PM

You can set k = 1 in SVM because the separating hyperplane is scale-invariant: multiplying w and b by any positive scalar α multiplies the functional margin by the same α without changing the classifier, so k can be fixed to 1 to remove this redundant degree of freedom and define a canonical hyperplane

What k means

- In the hard-margin setup, the parallel "margin" planes are defined by $w^\top x + b = +k$ and $w^\top x + b = -k$, and the decision hyperplane is $w^\top x + b = 0$; choosing k fixes the scale of w and b and therefore the numerical value of the functional margins $y_i(w^\top x_i + b)$. see.stanford +1
- When k is normalized to 1 (canonical form), support vectors satisfy $y_i(w^\top x_i + b) = 1$, the two margin planes become $w^\top x + b = \pm 1$, and the geometric margin (distance from the hyperplane to the closest points) is $1/\|w\|$, giving a total margin width $2/\|w\|$. math.uh +1

Why k can be 1

- The geometric margin is invariant to any common rescaling of the parameters (w, b); replacing (w, b) with $(\alpha w, \alpha b)$ leaves the classifier unchanged but scales the functional margins by α, so one is free to impose an arbitrary scaling constraint such as setting the smallest functional margin to 1. see.stanford
- This normalization is called the canonical hyperplane; formally, rescale (w, b) so the closest points satisfy $|\langle w, x_i \rangle + b| = 1$, i.e., $y_i(w^\top x_i + b) = 1$ for support vectors. math.uh

1. Hard Margin SVM

A **hard margin SVM (Support Vector Machine)** is the *purest* version of SVM — it assumes that your data is **perfectly linearly separable**.

Goal:
Find the hyperplane that *maximizes the margin* — the distance between the hyperplane and the nearest data points from each class — **without any misclassification**.

Mathematically, the hyperplane is:

$$w \cdot x + b = 0$$

Where:

- w is the normal vector to the hyperplane
- b is the bias (offset)
- x is a data point

Optimization Objective:
We want to:

$$\min_{w,b} \frac{1}{2}\|w\|^2$$

Subject to:

$$y_i(w \cdot x_i + b) \geq 1 \quad \forall i$$

Where $y_i \in \{-1, +1\}$.

The condition $y_i(w \cdot x_i + b) \geq 1$ ensures **correct classification** with a **margin of at least 1**.

Geometric Interpretation:

- The **margin** is $\frac{2}{\|w\|}$.
- Only points lying *exactly* on the margin boundaries are called **support vectors**.
- If data is *not perfectly separable*, hard margin SVM fails — it can't find any solution because constraints can't all be satisfied.

So: **Hard marain = zero tolerance for misclassification or noise.**

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2. Soft Margin SVM

In real-world data, perfect separability is rare — there's noise, overlap, outliers.
Soft margin SVM allows some violations of the margin constraints using **slack variables**.

Slack Variables:
Introduce $\xi_i \geq 0$ for each data point to allow misclassifications or margin violations.

Constraint becomes:

$$y_i(w \cdot x_i + b) \geq 1 - \xi_i$$

- $\xi_i = 0$: correctly classified and outside margin
- $0 < \xi_i < 1$: correctly classified but inside margin
- $\xi_i > 1$: misclassified

Objective Function:
We now minimize both the margin size and the total penalty for violations:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N} \xi_i$$

Where:

- $\frac{1}{2}\|w\|^2$ → maximize margin
- $C\sum \xi_i$ → penalize misclassifications
- $C > 0$ → **regularization parameter**

HINGE LOSS

Intuitive idea

An SVM doesn't just care whether a prediction is correct — it also cares *how confidently* it's correct.
Hinge loss penalizes points that:

1. Are misclassified, **and**
2. Are correctly classified but **too close to the margin** (i.e., not confidently correct).

So it's a *margin-based loss* rather than a pure classification loss.

Mathematical form

For a single data point (x_i, y_i), where $y_i \in \{-1, +1\}$:

$$L_i = \max(0, 1 - y_i(w \cdot x_i + b))$$

How it behaves

- If $y_i(w \cdot x_i + b) \geq 1$:
  The point is on the correct side **and outside the margin**.
  → Loss = 0
  (No penalty — we're happy.)
- If $0 < y_i(w \cdot x_i + b) < 1$:
  The point is correctly classified but **within the margin**.
  → Loss = $1 - y_i(w \cdot x_i + b)$
  (Small penalty for being too close to the boundary.)
- If $y_i(w \cdot x_i + b) \leq 0$:
  The point is **misclassified**.
  → Loss is large (≥ 1).
  (Hefty penalty.)

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Connection to the soft margin SVM objective

The soft-margin SVM optimization problem:

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N} \xi_i$$

is equivalent to:

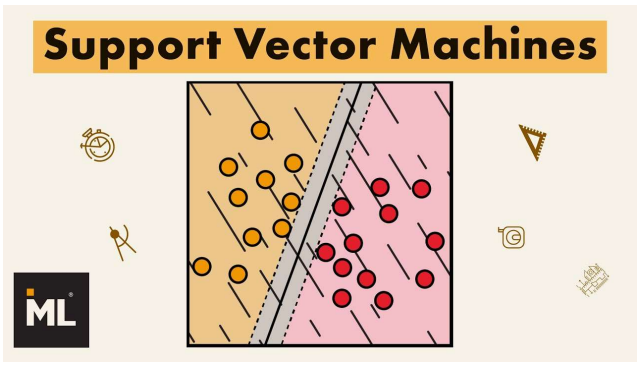
$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N} \max(0, 1 - y_i(w \cdot x_i + b))$$

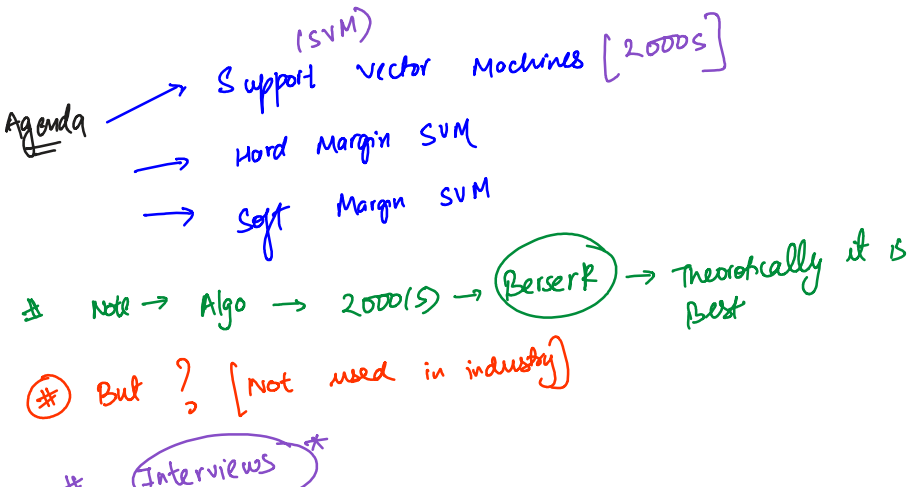
because $\xi_i = \max(0, 1 - y_i(w \cdot x_i + b))$ — the hinge loss itself.

In essence

- **Hinge loss = margin-based penalty.**
- It encourages:
  - **Large margin** (by pushing predictions away from the boundary)
  - **Few misclassifications**
- It's why SVMs produce *robust, maximum-margin* classifiers rather than just correct ones.







(*) Bu ... L...

Interviews *

SVM - 1
SVM - 2

→ Math Heavy algo → Mathematics

→ Thought Process [Idea]

Example

Problem Statement :

Imagine yourself as a Data Scientist at Google.

Task :

You've been asked to come up with model to classify emails as :

SPAM or HAM

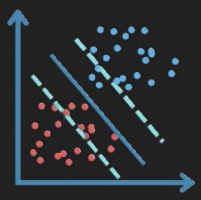


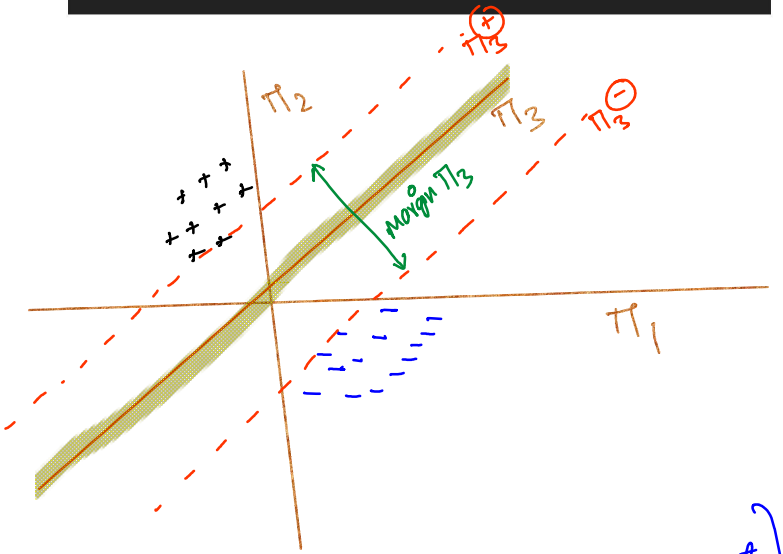
SVMs

Support Vector Machine

SVMs

- Popular in 2000's (late 90s)
- Kernel SVM
- Theoretically they are best
- Less frequently used nowadays
- Challenging Maths

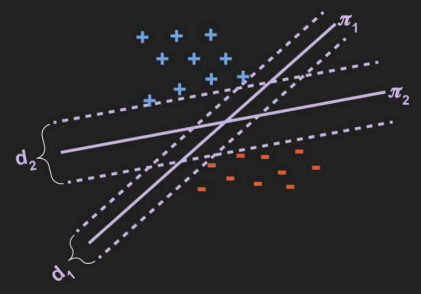




* [π_3 → Best → Because margin is highest]

Which line/hyperplane should we choose?

Imagine two hyperplanes π_1 and π_2



$\because$ Margin d_1 <<< Margin d_2

$\therefore$ Pick π_2 s.t. we get max/largest margin

Why do we need a large margin?

Because,

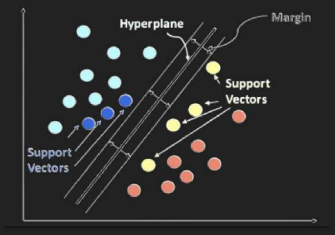
Larger the margin,

Better the separation

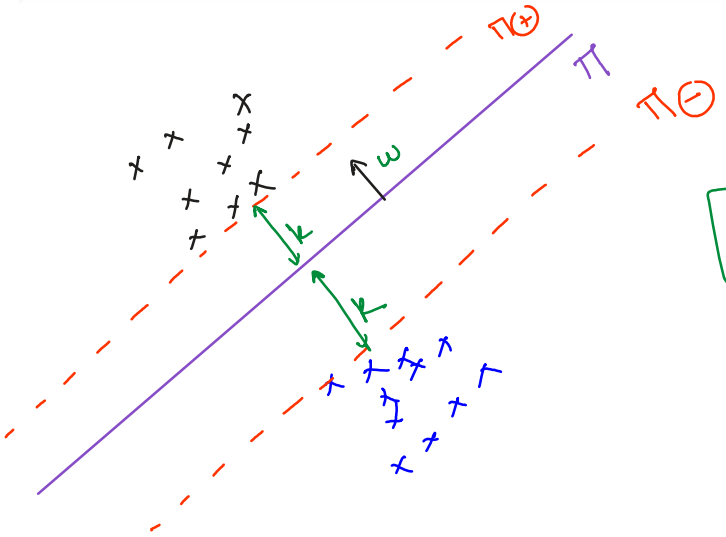
between the *+ve* and *-ve* samples.

Such classifiers where,

The margin should be as large as possible are called **margin-maximizing classifiers.**



NOTE: Distances from data points are measured perpendicular to any of the hyperplanes.



$$\pi \rightarrow w^T x + b = 0 \quad (*)$$

$$\pi^{(+)} \rightarrow w^T x + b = k$$

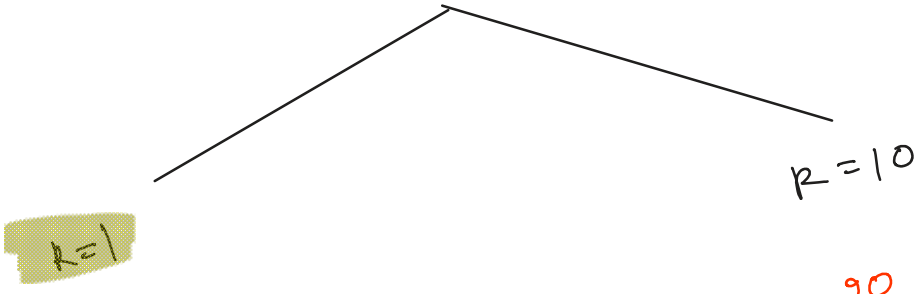
$$\pi^{(-)} \rightarrow w^T x + b = -k$$

$$\text{Margin}\left(\pi^{(+)}, \pi^{(-)}\right) = d\left(0, \pi^{(+)}\right) \rightarrow \frac{w^T(0) + b - k}{\|w\|}$$

$$d\left(0, \pi^{(-)}\right) \rightarrow \frac{w^T(0) + b + k}{\|w\|}$$

(Absolute distance) $\rightarrow \left| -d(0,\pi^{\oplus}) + d(0,\pi^{-}) \right|$

$\Rightarrow \left| \frac{(b+k)}{\|w\|} - \frac{(b-k)}{\|w\|} \right| = \frac{2k}{\|w\|}$



Margin $= \frac{2}{\|w\|}$

Margin $= \frac{20}{\|w\|}$

$w^T x = b + k$ *

$k=1 \rightarrow w^T x = b+1$

$k=20 \rightarrow \frac{w^T x}{20} = \frac{b+20}{20}$ — (II)

$\frac{w^T x}{20} = b' + 1$

Hard Margin SVM

(No misclassification allowed)

What do you mean by generalization in terms of SVM?

3 options

Active Duration(Most preferred: 30 seconds)

Appears for: 30 Secs

A. How far the hyperplane is from the training datapoints

B. How accurately the SVM can predict outcomes for unseen data

C. How accurately the SVM classifies training datapoints

- $\pi+ : w^T * x + b = 40$
- $\pi- : w^T * x + b = -50$

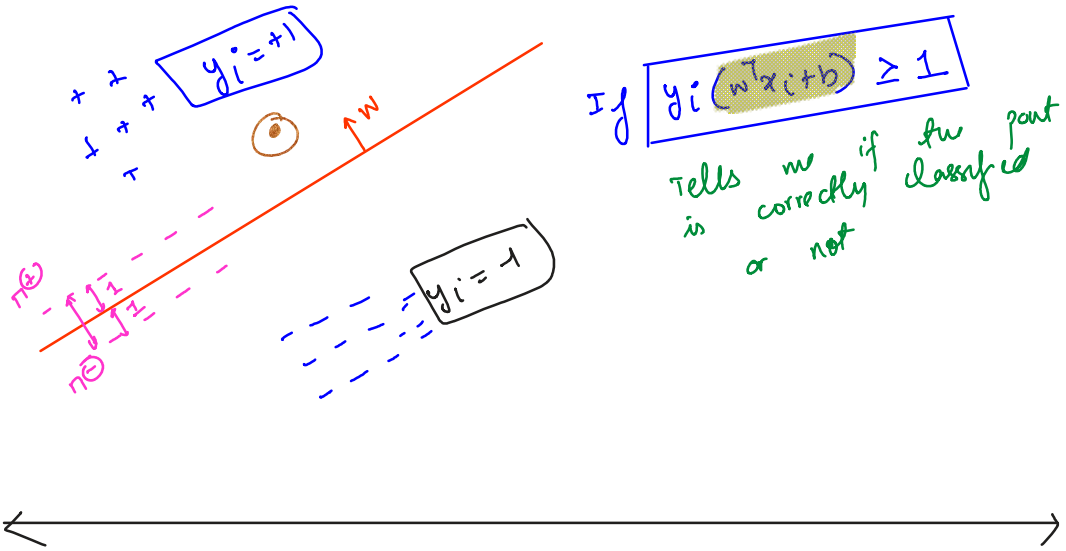
then margin will be:

$\frac{(40-b) - (-50-b)}{\|w\|} = \frac{90}{\|w\|}$

English → (hyp which max(margin) + no misclassification) *

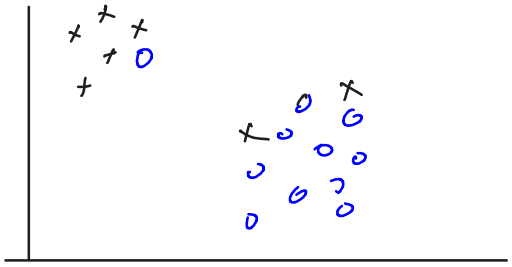
(2) such that $y_i(w^T x_i + b) \geq 1$ * i to N

Maths → max $\left(\frac{2}{\|w\|}\right)$ such that $y_i(w^T x_i + b) \geq 1$ $\forall i = 1$ to N



If $y_i(w^T x_i + b) \geq 1$

Tells me if the point is correctly classified or not



Why the $y_i (w^T x + b) \geq 1$ constraint?

Our goal is to:

- completely seperate the two classes
- with a margin as large as possible

With this constraint:

- all *+ve* data points should lie beyond π^+
- all *-ve* data points should lie beyond π^-

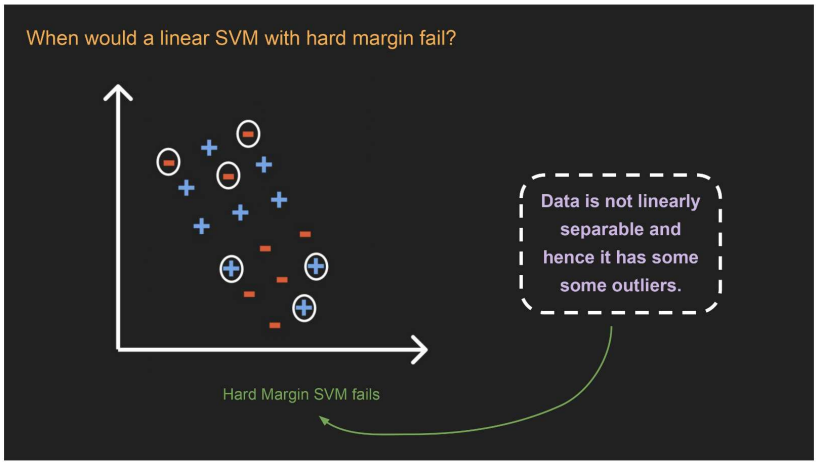
Which means, we strictly impose that:

All instances must be

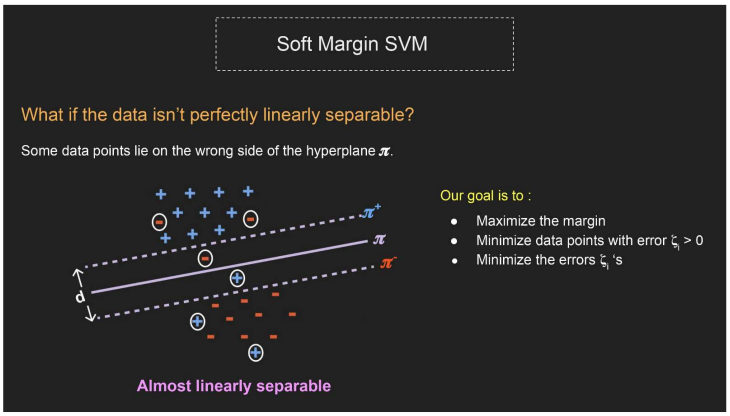
- off the margin and
- on the correct side
- i.e., Zero Error

This is called **Hard Margin Classification**



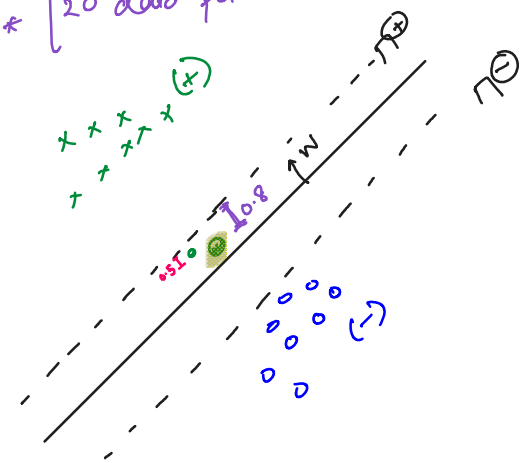


Soft Margin SVM → allow errors (Try to minimise them)



(#) Soft Margin SVM → we introduce ξ_i → slack variable

* [20 data points → 20 slack variables]



If point is correctly classified + well away from margin $\xi_i = 0$

ξ_i → measures how wrong is the data point

$\xi_i = 0$ or $\neq 0$

ξ_i → error = 0.8

[Hard margin SVM equation]

[Soft margin SVM equation]

max (margin) s.t minimum misclassification / total error

$$\max\left(\frac{2}{\|w\|}\right) \quad \text{s.t} \quad \left(w^T x_i + b\right) \geq 1 - \xi_i \quad \forall\ i = 1 \text{ to } N \qquad \xi_i \geq 0$$

$$\left[\min\left(\frac{\|w\|}{2}\right) + C\frac{\sum_{i=1}^{N}\xi_i}{N}\right] \text{ s.t } \left(w^T x_i + b\right) \geq 1 - \xi_i \quad \forall\ i = 1 \text{ to } N \qquad \xi_i \geq 0$$

C: Regularisation constant

In general,

$$y_i(w^T x_i + b) \geq 1 - \zeta_i$$

Note:

- $\zeta_i = 0$ for all correctly placed points
- $\zeta_i > 0$ for all misclassified points

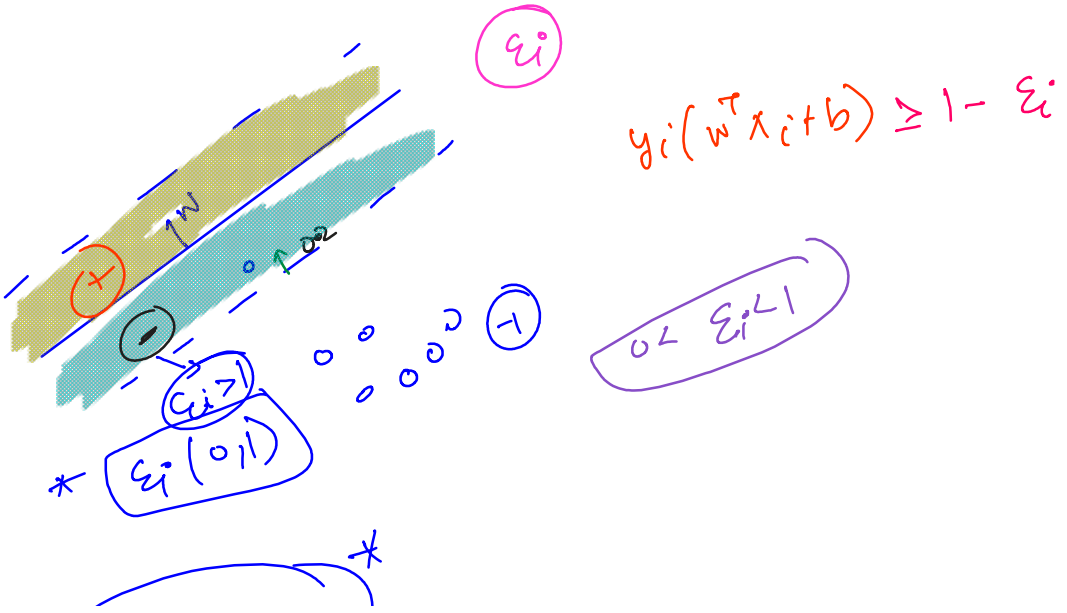
$\zeta_i \geq 0$ for all data points

Now, the optimization fn changes to :

$$min_{w,b}\ \frac{||w||}{2} + \frac{C}{N}\sum_{i=1}^{N}\zeta_i \quad s.t.\ (w^T x_i + b)y_i \geq 1 - \zeta_i$$

$$\forall\, i : 1 \rightarrow N \text{ and } \zeta_i \geq 0$$

This is known as Soft Margin SVM



$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

10:21 pm

Summary

Hard Margin (0 M.C)

Softmargin

Hard Margin SVM

Our goal is to maximize the margin
s.t. the +ve and -ve samples are on different sides of hyperplane π

Therefore, the margin constraints are :

For $y_i = +1 \longrightarrow \pi^+ = y_i (w^T x + b) \geq 1$

For $y_i = -1 \longrightarrow \pi^- = y_i (w^T x + b) \leq -1$

If we club both these equations:

$$argmax_{w,b}\left(\frac{2}{||w||}\right) \; s.t. \; y_i(w^T x_i + b) \geq 1 \quad \forall \; i : 1 \rightarrow N$$

Max Margin

No MC + Nothing in danger zone

(Soft margin SVM)

In general,

$$y_i(w^T x_i + b) \geq 1 - \zeta_i$$

Note:

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- $\zeta_i > 0$ for all misclassified points

$\rightarrow \; \zeta_i \geq 0$ for all data points

Now, the optimization fn changes to :

$$min_{w,b} \; \frac{||w||}{2} + \frac{C}{N}\sum_{i=1}^{N} \zeta_i \; \; s.t. \; (w^T x_i + b) y_i \geq 1 - \zeta_i$$

$$\forall \; i : 1 \rightarrow N \; and \; \zeta_i \geq 0$$

This is known as Soft Margin SVM

C → Regularisation Parameter

C ↑↑

$\sum_{i=1}^{N} 1.0$

C↑↑

$$\min \quad \frac{\|w\|}{2} + \frac{C}{N} \sum_{i=1}^{N} \xi_i$$

C → Hyperparameter in SVM

What's the use of 'C' here?

C is a hyperparameter.
It controls whether to focus on :

- Maximizing margin
- Minimizing errors ζ_i 's

Case 1

As C ↓

- Model becomes too lenient towards misclassifications.
- More importance is given to maximizing margins.
- The model may Underfit.

Case 2

As C ↑

- Model becomes less lenient towards misclassifications.
- More importance is given to minimizing errors.
- The model may Overfit.

$$\min \left(\text{Margin} + C(\text{misclassfc}^{n}) \right)$$

$C = 1000$

Soft margin SVM

$$\min \quad \frac{\|w\|}{2} + C \left(\frac{\sum_{i=1}^{N} \xi_i}{N} \right)$$

→ Hinge Loss

2

$\left(\frac{1}{\lambda}\right)$ *

Reg. param

misclassfc n *

C

margin

pt x_i

pt c.c t away from margin

$\Rightarrow y_i(w^T x_i + b) \geq 1$ $\Rightarrow$ $y_i f(x) \geq 1$ (Ideal)

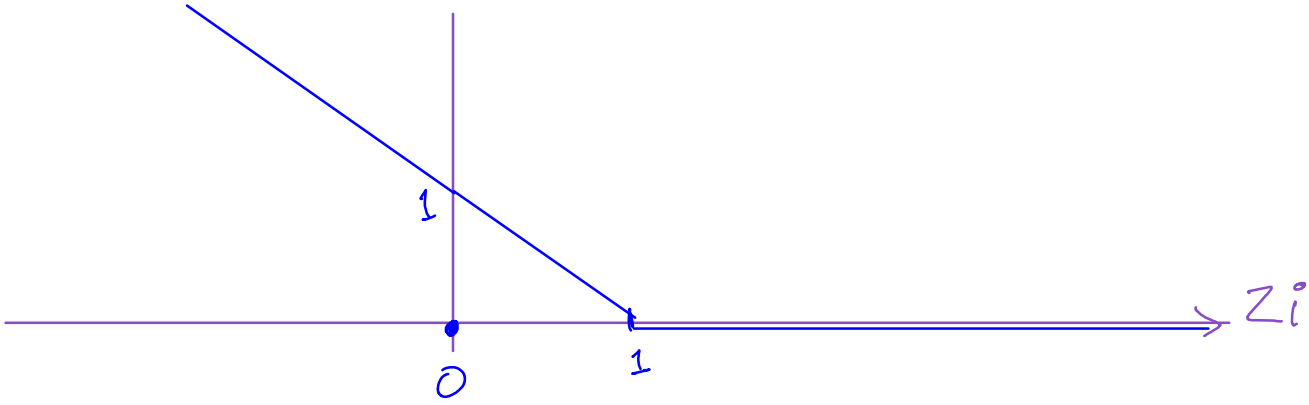
↙ Hinge loss

$y_i(w^T x_i + b) \geq 1 - \xi_i$

$\xi_i \geq 1 - y_i f(x_i)$

z_i

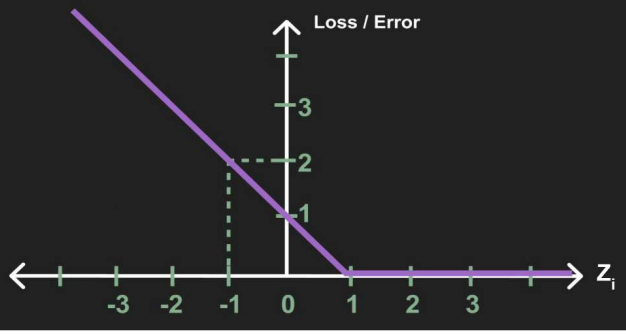
$\xi_i \geq 1 - y_i(w^T x_i + b)$



Intuition of Hinge Loss

Let's define $(w^T x + b)$ as $f(x_i)$

Assume X-axis to be : $Z_i = y_i(f(x_i)) = y_i(w^T x_i + b)$



$$\xi_i = \max(0, 1 - Z_i) \quad *$$

What can we conclude from this?

- Error cannot be negative. $\zeta_i \geq 0$
- According to the constraint -

  $y_i (w^T x + b) > -1 - \zeta$

  $= Z_i > 1 - \zeta$

  $= \zeta_i \geq 1 - \zeta - Z_i$



We can combine these two equations and get Hinge Loss $\zeta_i = max(0, 1 - Z_i)$

NOTE: As Z_i ↑ ζ_i ↓ till zero

As Z_i ↓ ζ_i ↑

Comparison with Log Loss

Logistic Regression

- $y_i \in \{0, 1\}$
- So, Log Loss =

$$\sum_{i=1}^{N} y_i log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)$$

Support Vector Machine

- $y_i \in \{-1, 1\}$
- So, Hinge loss = $\sum_{i=1}^{N} \zeta_i$

$$\sum_{i=1}^{N} max(0, 1 - Z_i)$$



(#) Does SVM affected by class imbalance ?

(#) complete assignment *